

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1711

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninfor solve problems effectively. (BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

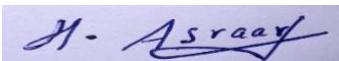
Analytical Problem Solving

Code of Conduct:

I H ASRAAR AHMED-192524028 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Analytical Problem Solving

1. Bridge and Torch Problem (State-space formulation and UCS)

Four people must cross a narrow bridge at night. They have one torch, and the bridge holds at most two people at a time. Anyone crossing must carry the torch, and a pair walks at the slower person's pace. Their crossing times are 1, 2, 5 and 10 minutes.

- i. Formulate the problem: state representation, initial state, actions, transition model, goal test and path cost.
- ii. Use Uniform Cost Search to find a schedule that gets everyone across in the minimum total time.
- iii. Explain why the obvious strategy (the fastest person escorts everyone) is not optimal.

ANSWER:

1. Problem Identification

The objective is to find the **minimum-cost sequence of actions** for four people to cross a bridge at night using one torch.

- Crossing times: **1, 2, 5, 10 minutes**
 - Maximum people on bridge: **2**
 - Torch must always be carried.
 - Two people crossing together take the time of the **slower person**.
 - Goal: **Everyone must reach the other side in minimum time.**
-

2. State Representation

A state represents:

(People on starting side, Torch position)

For example:

$$(\{1, 2, 5, 10\}, L)$$

means all four people are on the left side and the torch is on the left.

Initial state:

$$\boxed{(\{1, 2, 5, 10\}, L)}$$

Goal state:

$$\boxed{(\{\}, \{1, 2, 5, 10\}R)}$$

3. Initial State

Initially, all four people and the torch are on the starting side.

$$\boxed{S_0 = (\{1, 2, 5, 10\}, L)}$$

4. Operators / Actions

The possible actions are:

- Two people cross together.
- One person returns with the torch.

For two people A and B :

$$Cost(A, B) = \max(A, B)$$

Example:

$$(1, 10) \rightarrow = 10 \text{ minutes}$$

because the slower person takes 10 minutes.

5. Transition Model

After every action:

1. Selected person/persons move to the opposite side.
2. Torch moves to the opposite side.
3. The corresponding crossing time is added to the total cost.

Example:

$$(1, 2, 5, 10, L) \xrightarrow{1,2} (5, 10, R)$$

Cost = 2 minutes.

6. Goal Test

The goal is reached when **all people are on the opposite side**:

$$(\quad, \{ \quad \} R)$$

7. Path Cost

Path cost is the **total time taken for all crossings**.

$$g(n) = \sum \text{cost of each action}$$

Analytical problem solving aims to find the path with the **minimum total cost**.

2. 8. Uniform Cost Search

UCS expands the state with the **lowest accumulated cost** first.

The optimal path is:

Step	Action	Cost	Total
1	1 and 2 cross →	2	2
2	1 returns ←	1	3
3	5 and 10 cross →	10	13
4	2 returns ←	2	15
5	1 and 2 cross →	2	17

Therefore:

Minimum Cost = 17 minutes

Final Solution

(1, 2) → , 1 ← , (5, 10) → , 2 ← , (1, 2) →

Total:

$$2 + 1 + 10 + 2 + 2 = 17 \text{ minutes}$$

9. Analytical Problem-Solving Explanation

This problem demonstrates the following **analytical problem-solving steps**:

Problem → **State Representation** → **Initial State** → **Actions** → **Transition** → **Goal Test**
→ **Cost Calculation** → **Search** → **Optimal Solution**

The important idea is that we **analyze different possible states and actions** and select the sequence having the **minimum cost**.

Why the fastest-person strategy is not optimal?

If the 1-minute person escorts everyone:

$$10 + 1 + 5 + 1 + 2 = 19$$

So:

$$19 > 17$$

Hence, the fastest-person strategy is **not optimal**.

The analytical solution instead sends **5 and 10 together**, saving time:

$$17 \text{ minutes}$$

Conclusion: The Bridge and Torch problem is an example of **analytical problem solving using state-space representation and Uniform Cost Search to obtain an optimal solution**.

4. Route Finding with A* Search

A navigation app has to find the shortest route from S to G. The road costs are:

Edge	Cost	Edge	Cost
S–A	2	B–D	3
S–B	5	C–D	1
A–C	4	C–G	6
Edge	Cost	Edge	Cost
A–D	7	D–G	2

The heuristic values are $h(S)=7$, $h(A)=6$, $h(B)=4$, $h(C)=3$, $h(D)=2$ and $h(G)=0$. i.

Apply A* search and show the open and closed lists at each step.

iii. Give the path found and its cost. iii. Check whether the heuristic is admissible and consistent. iv. Apply Greedy Best-First Search to the same graph and compare the results.

ANSWER:

Given Graph

Edges and costs:

- $S \rightarrow A = 2$
- $S \rightarrow B = 5$
- $A \rightarrow C = 4$
- $A \rightarrow D = 7$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 6$

- $D \rightarrow G = 2$

Heuristic values:

Node	$h(n)$
S	7
A	6
B	4
C	3
D	2
G	0

i. Apply A* Search

A* uses:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ = actual cost from S to n
- $h(n)$ = estimated cost from n to G
- $f(n)$ = total estimated cost

Step 0

Initially:

OPEN:

Node	g	h	f
S	0	7	7

CLOSED: —

Select **S** because it has the lowest $f = 7$.

Step 1 – Expand S

From S:

- $S \rightarrow A: g = 2$

$$f(A) = 2 + 6 = 8$$

- $S \rightarrow B: g = 5$

$$f(B) = 5 + 4 = 9$$

OPEN:

Node	g	h	f
A	2	6	8
B	5	4	9

CLOSED:

$\{S\}$

Select **A**.

Step 2 – Expand A

From A:

A → **C**

$$\begin{aligned}g(C) &= 2 + 4 = 6 \\f(C) &= 6 + 3 = 9\end{aligned}$$

A → **D**

$$\begin{aligned}g(D) &= 2 + 7 = 9 \\f(D) &= 9 + 2 = 11\end{aligned}$$

Now:

OPEN:

Node	g	h	f
B	5	4	9
C	6	3	9
D	9	2	11

CLOSED:

$\{S, A\}$

There is a tie between B and C. We can expand **B first**.

Step 3 – Expand B

From B:

$$B \rightarrow D$$

New cost:

$$g(D) = g(B) + 3 = 5 + 3 = 8$$

Therefore:

$$f(D) = 8 + 2 = 10$$

This is better than the previous $g(D) = 9$, so update D.

OPEN:

Node	g	h	f
C	6	3	9
D	8	2	10

CLOSED:

$$\{S, A, B\}$$

Select **C** because $f(C) = 9$.

Step 4 – Expand C

From C:

C → D

$$\begin{aligned}g(D) &= 6 + 1 = 7 \\f(D) &= 7 + 2 = 9\end{aligned}$$

This is better than the previous $g(D) = 8$, so update D.

C → G

$$\begin{aligned}g(G) &= 6 + 6 = 12 \\f(G) &= 12 + 0 = 12\end{aligned}$$

OPEN:

Node	g	h	f
D	7	2	9
G	12	0	12

CLOSED:

$$\{S, A, B, C\}$$

Select **D**.

Step 5 – Expand D

From D:

$$\begin{aligned}D &\rightarrow G \\g(G) &= 7 + 2 = 9\end{aligned}$$

$$f(G) = 9 + 0 = 9$$

This improves G from 12 to 9.

OPEN:

Node	g	h	f
G	9	0	9

CLOSED:

$$\{S, A, B, C, D\}$$

Select G.

Since **G is the goal**, stop.

3. ii. Path Found and Cost

Tracing the parent nodes:

$$G \leftarrow D \leftarrow C \leftarrow A \leftarrow S$$

Therefore:

$$\boxed{S \rightarrow A \rightarrow C \rightarrow D \rightarrow G}$$

Cost:

$$2 + 4 + 1 + 2 = \boxed{9}$$

Answer:

Shortest path = $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$

Minimum cost = 9

4. iii. Check Admissibility and Consistency

Admissibility

A heuristic is **admissible** if it never overestimates the actual minimum cost to the goal.

Actual shortest costs to G:

- $h^*(G) = 0$
- $h^*(D) = 2$
- $h^*(C) = 1 + 2 = 3$
- $h^*(A) = 4 + 1 + 2 = 7$
- $h^*(B) = 3 + 2 = 5$
- $h^*(S) = 9$

Compare with given heuristic:

N o d e	G i v e n h (n)	Act ual mi ni mu m	Admi ssible ?
S	7	9	✓
A	6	7	✓
B	4	5	✓
C	3	3	✓
D	2	2	✓
G	0	0	✓

Therefore:

Heuristic is admissible

Consistency

For every edge $n \rightarrow n'$, a consistent heuristic must satisfy:

$$h(n) \leq c(n, n') + h(n')$$

Check:

- $S \rightarrow A: 7 \leq 2 + 6 = 8 \checkmark$

- $S \rightarrow B: 7 \leq 5 + 4 = 9\checkmark$
- $A \rightarrow C: 6 \leq 4 + 3 = 7\checkmark$
- $A \rightarrow D: 6 \leq 7 + 2 = 9\checkmark$
- $B \rightarrow D: 4 \leq 3 + 2 = 5\checkmark$
- $C \rightarrow D: 3 \leq 1 + 2 = 3\checkmark$
- $C \rightarrow G: 3 \leq 6 + 0 = 6\checkmark$
- $D \rightarrow G: 2 \leq 2 + 0 = 2\checkmark$

Therefore:

Heuristic is consistent

5. iv. Greedy Best-First Search

Greedy Best-First Search uses **only the heuristic**:

$f(n) = h(n)$

It ignores the actual cost $g(n)$.

Step 1

From S:

- A: $h = 6$
- B: $h = 4$

Greedy chooses **B** because:

$$4 < 6$$

Step 2

From B:

$$B \rightarrow D$$

D has:

$$h(D) = 2$$

So choose D.

Step 3

From D:

$$\begin{aligned} D &\rightarrow G \\ h(G) &= 0 \end{aligned}$$

Goal reached.

Therefore Greedy finds:

$$\boxed{S \rightarrow B \rightarrow D \rightarrow G}$$

Cost:

$$5 + 3 + 2 = \boxed{10}$$

A* vs Greedy Best-First Search

Feature	A* Search	Greedy Best-First
Evaluation	$f = g + h$	$f = h$
Considers actual cost?	Yes	No
Path found	$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	$S \rightarrow B \rightarrow D \rightarrow G$
Cost	9	10
Optimal?	Yes	Not necessarily
Main idea	Balance cost + heuristic	Follow apparently closest node

Final Conclusion

This problem demonstrates **analytical problem solving using informed search**.

A* gives the optimal route with cost 9

while

Greedy Best-First gives a route with cost 10

The reason A* performs better is that it considers **both the actual cost $g(n)$ and estimated cost $h(n)$** , whereas Greedy Best-First considers only $h(n)$.

5. 8-Puzzle with Heuristic Search

Solve the 8-puzzle from the start state below to the goal state, where _ is the blank.

Start				Goal		
2	8	3		1	2	3
1	6	4		8	—	4
7	—	5		7	6	5

- Formulate the problem components.
- Define h_1 (number of misplaced tiles) and h_2 (sum of Manhattan distances), and compute both for the start state.
- Solve the puzzle with A* using h_2 and draw the search tree.
- Explain why h_2 dominates h_1 .

ANSWER:

Start State

2	8	3
1	6	4
7	—	5

Goal State

1	2	3
8	—	4
7	6	5

i. Problem Formulation

1. Initial State

2	8	3
1	6	4
7	—	5

2. Goal State

1	2	3
8	-	4
7	6	5

3. State Representation

A state is represented by the **3 × 3 arrangement of tiles**, including the blank.

4. Operators / Actions

The blank space can move:

- Up
- Down
- Left
- Right

provided the movement is within the puzzle boundary.

5. Transition Model

When the blank moves, it **swaps positions with the adjacent tile**.

Each movement has a cost of:

1

6. Goal Test

Check whether the current arrangement is equal to the goal state.

7. Path Cost

Total path cost = number of tile movements.

A* uses:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ = moves made so far
 - $h(n)$ = estimated moves remaining
 - $f(n)$ = estimated total cost
-

6. ii. Heuristics h_1 and h_2

h_1 : Number of Misplaced Tiles

h_1 counts the tiles that are **not in their goal positions**.

Start:

2	8	3
1	6	4
7	_	5

Goal:

1	2	3
8	_	4
7	6	5

Compare:

- 2 → misplaced ✓
- 8 → misplaced ✓
- 3 → correct
- 1 → misplaced ✓
- 6 → misplaced ✓
- 4 → correct
- 7 → correct
- 5 → correct

Therefore:

$$h_1 = 4$$

The blank is not counted.

h_2 : Manhattan Distance

Manhattan distance is:

$$|r_1 - r_2| + |c_1 - c_2|$$

for every tile.

Tile	Start Position	Goal Position	Distance
1	(2,1)	(1,1)	1

Tile	Start Position	Goal Position	Distance
2	(1,1)	(1,2)	1
3	(1,3)	(1,3)	0
4	(2,3)	(2,3)	0
5	(3,3)	(3,3)	0
6	(2,2)	(3,2)	1
7	(3,1)	(3,1)	0
8	(1,2)	(2,1)	2

Therefore:

$$h_2 = 1 + 1 + 0 + 0 + 0 + 1 + 0 + 2$$

$$\boxed{h_2 = 5}$$

So initially:

$$\boxed{h_1 = 4, h_2 = 5}$$

7. iii. A* Search Using h_2

A* formula:

$$\boxed{f(n) = g(n) + h_2(n)}$$

The optimal solution requires **5 moves**.

Step 0 — Start

$$\begin{array}{ccc} 2 & 8 & 3 \\ 1 & 6 & 4 \\ 7 & - & 5 \\ g = 0, h = 5, f = 5 \end{array}$$

The blank can move **Up, Left or Right**.

A* selects the promising state with the lowest f .

Step 1

Move blank **Up**:

$$\begin{array}{ccc} 2 & 8 & 3 \\ 1 & - & 4 \\ 7 & 6 & 5 \\ g = 1, h = 4 \\ f = 1 + 4 = \boxed{5} \end{array}$$

Step 2

Move blank **Left**:

$$\begin{array}{ccc} 2 & - & 3 \\ 1 & 8 & 4 \\ 7 & 6 & 5 \\ g = 2, h = 3 \\ f = 2 + 3 = \boxed{5} \end{array}$$

Step 3

Move blank **Up**:

$$\begin{array}{ccc}
 & 2 & 3 \\
 1 & 8 & 4 \\
 7 & 6 & 5 \\
 g = 3, h = 2 \\
 f = 3 + 2 = \boxed{5}
 \end{array}$$

Step 4

Move blank **Down**:

$$\begin{array}{ccc}
 1 & 2 & 3 \\
 & 8 & 4 \\
 7 & 6 & 5 \\
 g = 4, h = 1 \\
 f = 4 + 1 = \boxed{5}
 \end{array}$$

Step 5 — Goal

Move blank **Right**:

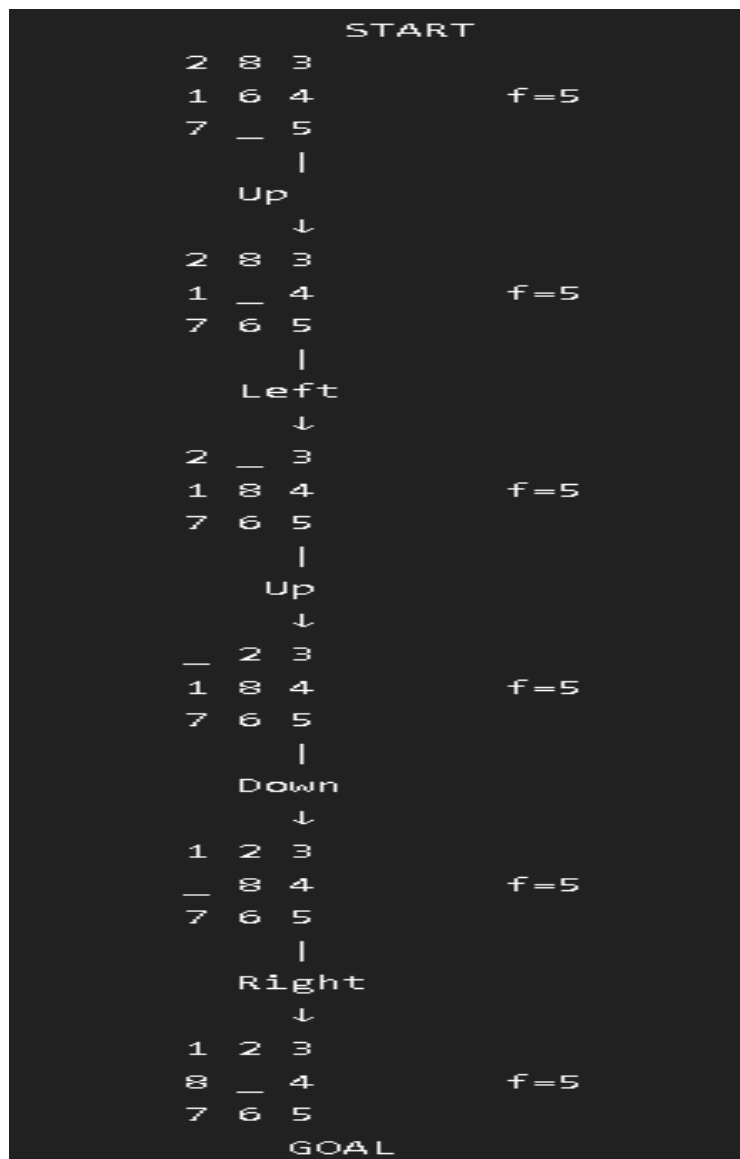
$$\begin{array}{|c|c|c|}
 \hline
 1 & 2 & 3 \\
 8 & & 4 \\
 7 & 6 & 5 \\
 \hline
 \end{array}$$

$$\begin{array}{l}
 g = 5, h = 0 \\
 f = 5 + 0 = \boxed{5}
 \end{array}$$

Goal reached.

8. A* Search Tree

The main A* search path can be shown as:



A* Path

Therefore, the sequence of blank movements is:

$$U \rightarrow L \rightarrow U \rightarrow D \rightarrow R$$

Number of moves:

5

So the minimum path cost is:

5

9. iv. Why Does h_2 Dominate h_1 ?

We have:

$$h_1(n) = \text{number of misplaced tiles}$$

and

$$h_2(n) = \text{sum of Manhattan distances}$$

For every state:

$$\boxed{h_2(n) \geq h_1(n)}$$

because every misplaced tile must move **at least one step**, while Manhattan distance tells us **how many row/column steps are actually required**.

For the starting state:

$$\begin{aligned} h_1 &= 4 \\ h_2 &= 5 \end{aligned}$$

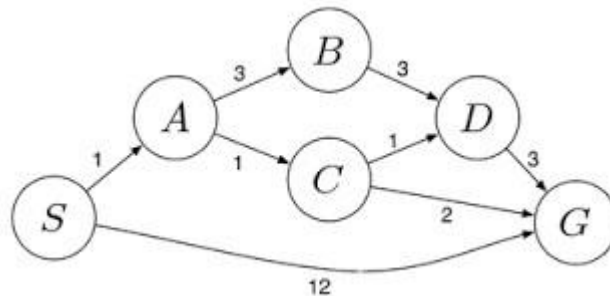
Therefore:

$$\boxed{h_2 > h_1}$$

Hence, h_2 gives more information about the actual remaining cost and **dominates** h_1 .

15. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



ANSWER:

Problem Formulation

State representation:

Each warehouse is represented as a state/node.

Initial state:

S

Goal state:

G

Actions:

Move from one warehouse to a directly connected warehouse.

Transition model:

Moving along an edge transfers the package from one warehouse to another with the cost shown on the edge.

Path cost:

Total cost of all routes travelled.

For UCS:

$$g(n) = \text{total cost from } S \text{ to node } n$$

Goal test:

Check whether the selected node is *G*.

2. Given Graph

The routes are:

Route Cost

$S \rightarrow A$ 1

$S \rightarrow G$ 12

$A \rightarrow B$ 3

Route Cost

A → C 1

B → D 3

C → D 1

C → G 2

D → G 3

3. Uniform Cost Search

UCS always expands the node having the **lowest path cost** $g(n)$.

Step 1: Start at S

$$g(S) = 0$$

OPEN:

[A, : 1 G: 12]

CLOSED:

[S]

Select **A**, because it has the lowest cost 1.

Step 2: Expand A

From A:

- A → B = 1 + 3 = 4
- A → C = 1 + 1 = 2

OPEN:

[C, :, 2 B: 4 G: 12]

CLOSED:

[S, A]

Select **C**, because $g(C) = 2$.

Step 3: Expand C

From C:

- C → D = 2 + 1 = 3
- C → G = 2 + 2 = 4

OPEN:

[D, :, 3 B: 4 G: 4]

CLOSED:

[S, A, C]

Select **D**, because $g(D) = 3$.

Step 4: Expand D

From D:

$$D \rightarrow G = 3 + 3 = 6$$

But G is already in OPEN with cost 4, so we keep the cheaper value.

OPEN:

$$[B, : 4 \ G: 4]$$

CLOSED:

$$[S, A, C, D]$$

Now G is selected with cost 4.

Step 5: Goal reached

G

Therefore, UCS terminates.

4. UCS Search Table

Step Expanded Node OPEN (lowest cost first) CLOSED

1	S	A:1, G:12	S
2	A	C:2, B:4, G:12	S, A
3	C	D:3, B:4, G:4	S, A, C
4	D	B:4, G:4	S, A, C, D
5	G	—	S, A, C, D, G

5. Optimal Path

Tracing the lowest-cost path:

$$\boxed{S \rightarrow A \rightarrow C \rightarrow G}$$

Cost:

$$1 + 1 + 2 = \boxed{4}$$

Final Answer

Least-cost delivery path:

$$\boxed{S \rightarrow A \rightarrow C \rightarrow G}$$

Minimum total cost:

$$\boxed{4}$$

Why UCS gives the optimal solution

UCS always expands the node with the **smallest accumulated path cost**. Since all route costs are positive, the first time the goal G is removed from OPEN, its path is the **minimum-cost path**.

Conclusion:

Thus, using **Uniform Cost Search**, the logistics company should transport the package through $S \rightarrow A \rightarrow C \rightarrow G$, with a minimum cost of 4.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem